File Edit View kali@kali: ~/go/network-lab/exp1/buffer-overflow-vulnerable

```
┌──(kali㉿kali)-[~/go/network-lab/exp1/buffer-overflow-vulnerable]
└─$ gosec ./...
[gosec] 2026/03/02 07:54:53 Including rules: default
[gosec] 2026/03/02 07:54:53 Excluding rules: default
[gosec] 2026/03/02 07:54:53 Including analyzers: default
[gosec] 2026/03/02 07:54:53 Excluding analyzers: default
[gosec] 2026/03/02 07:54:53 Import directory: /home/kali/go/network-lab/exp1/buffer-overflow-vulnerable
[gosec] 2026/03/02 07:54:53 Checking package: main
[gosec] 2026/03/02 07:54:53 Checking file: /home/kali/go/network-lab/exp1/buffer-overflow-vulnerable/buffer-overflow-vulnerable.go
Results:


[/home/kali/go/network-lab/exp1/buffer-overflow-vulnerable/buffer-overflow-vulnerable.go:23] - G103 (CWE-242): Use of unsafe calls should be audited (Confidence: HIGH, Severity: LOW)
    22:         for i := 0; i < len(data); i++ {
  > 23:                 *(*byte)(unsafe.Pointer(uintptr(ptr) + uintptr(i))) = data[i]
    24:         }

Autofix:

[/home/kali/go/network-lab/exp1/buffer-overflow-vulnerable/buffer-overflow-vulnerable.go:16] - G103 (CWE-242): Use of unsafe calls should be audited (Confidence: HIGH, Severity: LOW)
    15:         // UNSAFE pointer to first byte of struct
  > 16:         ptr := unsafe.Pointer(user)
    17:

Autofix:

[/home/kali/go/network-lab/exp1/buffer-overflow-vulnerable/buffer-overflow-vulnerable.go:37] - G104 (CWE-703): Errors unhandled (Confidence: HIGH, Severity: LOW)
    36:         fmt.Print("Enter username: ")
  > 37:         fmt.Scanln(&input)
    38:

Autofix:

Summary:
  Gosec  : dev
  Files  : 1
  Lines  : 43
  Nosec  : 0
  Issues : 3


┌──(kali㉿kali)-[~/go/network-lab/exp1/buffer-overflow-vulnerable]
└─$
```

```
┌──(kali㉿kali)-[~/go/network-lab/exp1/buffer-overflow-fixed]
└─$ gosec ./...
[gosec] 2026/03/02 07:58:33 Including rules: default
[gosec] 2026/03/02 07:58:33 Excluding rules: default
[gosec] 2026/03/02 07:58:33 Including analyzers: default
[gosec] 2026/03/02 07:58:33 Excluding analyzers: default
[gosec] 2026/03/02 07:58:33 Import directory: /home/kali/go/network-lab/exp1/buffer-overflow-fixed
[gosec] 2026/03/02 07:58:33 Checking package: main
[gosec] 2026/03/02 07:58:33 Checking file: /home/kali/go/network-lab/exp1/buffer-overflow-fixed/buffer-overflow-fixed.go
Results:


[/home/kali/go/network-lab/exp1/buffer-overflow-fixed/buffer-overflow-fixed.go:36] - G104 (CWE-703): Errors unhandled (Confidence: HIGH, Severity: LOW)
    35:         fmt.Print("Enter username: ")
  > 36:         fmt.Scanln(&input)
    37:

Autofix:

Summary:
  Gosec  : dev
  Files  : 1
  Lines  : 42
  Nosec  : 0
  Issues : 1


┌──(kali㉿kali)-[~/go/network-lab/exp1/buffer-overflow-fixed]
└─$
```

```
  GNU nano 8.6
package main

import (
        "fmt"
)

// fixed 16 byte buffer
type User struct {
        name  [16]byte
        admin bool
}

// SAFE write
func writeInputSafe(input string, user *User) {

        // reject overly long input
        if len(input) > len(user.name) {
                fmt.Println("Input too long! Max 16 characters allowed.")
                return
        }

        // copy safely (cannot overflow)
        copy(user.name[:], input)
}

func main() {
        user := User{
                name:  [16]byte{},
                admin: false,
        }

        fmt.Println("Admin before:", user.admin)

        var input string
        fmt.Print("Enter username: ")
        fmt.Scanln(&input)

        writeInputSafe(input, &user)

        fmt.Println("Stored name:", string(user.name[:]))
        fmt.Println("Admin after:", user.admin)
}
```

```
  GNU nano 8.6
package main

import (
        "fmt"
        "unsafe"
)

// fixed 16 byte buffer
type User struct {
        name  [16]byte
        admin bool
}

func writeInput(input string, user *User) {
        // UNSAFE pointer to first byte of struct
        ptr := unsafe.Pointer(user)

        // convert string -> byte slice
        data := []byte(input)

        // copy WITHOUT bounds checking (danger!)
        for i := 0; i < len(data); i++ {
                *(*byte)(unsafe.Pointer(uintptr(ptr) + uintptr(i))) = data[i]
        }
}

func main() {
        user := User{
                name:  [16]byte{},
                admin: false,
        }

        fmt.Println("Admin before:", user.admin)

        var input string
        fmt.Print("Enter username: ")
        fmt.Scanln(&input)

        writeInput(input, &user)

        fmt.Println("Stored name:", string(user.name[:]))
        fmt.Println("Admin after:", user.admin)
}
```

```
GNU nano 8.6
package main

import (
        "fmt"
        "os/exec"
)

func listFiles(userInput string) {
        // DANGEROUS: shell interpretation
        cmd := exec.Command("sh", "-c", "ls " + userInput)

        output, err := cmd.CombinedOutput()
        if err != nil {
                fmt.Println("Error:", err)
        }

        fmt.Println(string(output))
}

func main() {
        var dir string
        fmt.Print("Enter directory: ")
        fmt.Scanln(&dir)

        listFiles(dir)
}
```

```
GNU nano 8.6
package main

import (
        "fmt"
        "os/exec"
)

func listFilesSafe(dir string) {

        // SAFE: no shell involved
        cmd := exec.Command("ls", dir)

        output, err := cmd.CombinedOutput()
        if err != nil {
                fmt.Println("Error:", err)
                return
        }

        fmt.Println(string(output))
}

func main() {
        var dir string
        fmt.Print("Enter directory: ")
        fmt.Scanln(&dir)

        listFilesSafe(dir)
}
```

```
┌──(kali㉿kali)-[~/go/network-lab/exp1/unsafe-os-commands-vulnerable]
└─$ gosec ./...
[gosec] 2026/03/02 08:02:40 Including rules: default
[gosec] 2026/03/02 08:02:40 Excluding rules: default
[gosec] 2026/03/02 08:02:40 Including analyzers: default
[gosec] 2026/03/02 08:02:40 Excluding analyzers: default
[gosec] 2026/03/02 08:02:40 Import directory: /home/kali/go/network-lab/exp1/unsafe-os-commands-vulnerable
[gosec] 2026/03/02 08:02:40 Checking package: main
[gosec] 2026/03/02 08:02:40 Checking file: /home/kali/go/network-lab/exp1/unsafe-os-commands-vulnerable/main.go
Results:


[/home/kali/go/network-lab/exp1/unsafe-os-commands-vulnerable/main.go:10] - G204 (CWE-78): Subprocess launched with a potential tainted input or cmd arguments (Confidence: HIGH, Severity: MEDIUM)
    9:  // DANGEROUS: shell interpretation
  > 10:         cmd := exec.Command("sh", "-c", "ls " + userInput)
    11:

Autofix:

[/home/kali/go/network-lab/exp1/unsafe-os-commands-vulnerable/main.go:23] - G104 (CWE-703): Errors unhandled (Confidence: HIGH, Severity: LOW)
    22:         fmt.Print("Enter directory: ")
  > 23:         fmt.Scanln(&dir)
    24:

Autofix:

Summary:
  Gosec  : dev
  Files  : 1
  Lines  : 27
  Nosec  : 0
  Issues : 2
```

```
┌──(kali㉿kali)-[~/go/network-lab/exp1/unsafe-os-commands-fixed]
└─$ gosec ./...
[gosec] 2026/03/02 08:03:55 Including rules: default
[gosec] 2026/03/02 08:03:55 Excluding rules: default
[gosec] 2026/03/02 08:03:55 Including analyzers: default
[gosec] 2026/03/02 08:03:55 Excluding analyzers: default
[gosec] 2026/03/02 08:03:55 Import directory: /home/kali/go/network-lab/exp1/unsafe-os-commands-fixed
[gosec] 2026/03/02 08:03:55 Checking package: main
[gosec] 2026/03/02 08:03:55 Checking file: /home/kali/go/network-lab/exp1/unsafe-os-commands-fixed/main.go
Results:


[/home/kali/go/network-lab/exp1/unsafe-os-commands-fixed/main.go:11] - G204 (CWE-78): Subprocess launched with variable (Confidence: HIGH, Severity: MEDIUM)
    10:         // SAFE: no shell involved
  > 11:         cmd := exec.Command("ls", dir)
    12:

Autofix:

[/home/kali/go/network-lab/exp1/unsafe-os-commands-fixed/main.go:25] - G104 (CWE-703): Errors unhandled (Confidence: HIGH, Severity: LOW)
    24:         fmt.Print("Enter directory: ")
  > 25:         fmt.Scanln(&dir)
    26:

Autofix:

Summary:
  Gosec  : dev
  Files  : 1
  Lines  : 29
  Nosec  : 0
  Issues : 2
```

```
GNU nano 8.6
package main

import "fmt"

func logMessage(userInput string) {
	// DANGEROUS: user controls format string
	fmt.Printf(userInput)
	fmt.Println()
}

func main() {
	var name string
	fmt.Print("Enter your name: ")
	fmt.Scanln(&name)

	logMessage(name)
}
```

```
package main

import "fmt"

func logMessageSafe(userInput string) {
	fmt.Printf("%s\n", userInput) // SAFE
}

func main() {
	var name string
	fmt.Print("Enter your name: ")
	fmt.Scanln(&name)

	logMessageSafe(name)
}
```

```
┌──(kali㉿kali)-[~/go/network-lab/exp1/misused-format-specifiers-vulnerable]
└─$ gosec ./...
[gosec] 2026/03/02 08:05:44 Including rules: default
[gosec] 2026/03/02 08:05:44 Excluding rules: default
[gosec] 2026/03/02 08:05:44 Including analyzers: default
[gosec] 2026/03/02 08:05:44 Excluding analyzers: default
[gosec] 2026/03/02 08:05:44 Import directory: /home/kali/go/network-lab/exp1/misused-format-specifiers-vulnerable
[gosec] 2026/03/02 08:05:44 Checking package: main
[gosec] 2026/03/02 08:05:44 Checking file: /home/kali/go/network-lab/exp1/misused-format-specifiers-vulnerable/main.go
Results:


[/home/kali/go/network-lab/exp1/misused-format-specifiers-vulnerable/main.go:14] - G104 (CWE-703): Errors unhandled (Confidence: HIGH, Severity: LOW)
    13:         fmt.Print("Enter your name: ")
  > 14:         fmt.Scanln(&name)
    15:

Autofix:

Summary:
  Gosec  : dev
  Files  : 1
  Lines  : 18
  Nosec  : 0
  Issues : 1
```

```
┌──(kali㉿kali)-[~/go/network-lab/exp1/misused-format-specifiers-fixed]
└─$ gosec ./...
[gosec] 2026/03/02 08:07:13 Including rules: default
[gosec] 2026/03/02 08:07:13 Excluding rules: default
[gosec] 2026/03/02 08:07:13 Including analyzers: default
[gosec] 2026/03/02 08:07:13 Excluding analyzers: default
[gosec] 2026/03/02 08:07:13 Import directory: /home/kali/go/network-lab/exp1/misused-format-specifiers-fixed
[gosec] 2026/03/02 08:07:14 Checking package: main
[gosec] 2026/03/02 08:07:14 Checking file: /home/kali/go/network-lab/exp1/misused-format-specifiers-fixed/main.go
Results:


Summary:
  Gosec  : dev
  Files  : 1
  Lines  : 15
  Nosec  : 0
  Issues : 0
```

```
GNU nano 8.6
package main

import "fmt"

// intentionally vulnerable
const dbPassword = "MyDBPassword123!"
const apiKey = "sk_live_51H8dExampleSecretKey"
const jwtSecret = "very-secret-signing-key"

func main() {
	fmt.Println("Connecting using:", dbPassword, apiKey, jwtSecret)
}
```

```
┌──(kali㉿kali)-[~/go/network-lab/exp1/hardcooded-credentials-vulnerable]
└─$ gosec ./...
[gosec] 2026/03/02 08:09:49 Including rules: default
[gosec] 2026/03/02 08:09:49 Excluding rules: default
[gosec] 2026/03/02 08:09:49 Including analyzers: default
[gosec] 2026/03/02 08:09:49 Excluding analyzers: default
[gosec] 2026/03/02 08:09:49 Import directory: /home/kali/go/network-lab/exp1/hardcooded-credentials-vulnerable
[gosec] 2026/03/02 08:09:49 Checking package: main
[gosec] 2026/03/02 08:09:49 Checking file: /home/kali/go/network-lab/exp1/hardcooded-credentials-vulnerable/main.go
Results:


[/home/kali/go/network-lab/exp1/hardcooded-credentials-vulnerable/main.go:7] - G101 (CWE-798): Potential hardcoded credentials (Confidence: LOW, Severity: HIGH)
    6: const dbPassword = "MyDBPassword123!"
  > 7: const apiKey = "sk_live_51H8dExampleSecretKey"
    8: const jwtSecret = "very-secret-signing-key"

Autofix:

Summary:
  Gosec  : dev
  Files  : 1
  Lines  : 12
  Nosec  : 0
  Issues : 1
```

```
  GNU nano 8.6
package main

import (
        "fmt"
        "os"

        "golang.org/x/crypto/bcrypt"
)

func verifyPassword(hash string, password string) bool {
        err := bcrypt.CompareHashAndPassword([]byte(hash), []byte(password))
        return err == nil
}

func login(user, pass string) bool {
        adminUser := os.Getenv("APP_ADMIN_USER")
        adminHash := os.Getenv("APP_ADMIN_HASH")

        if user != adminUser {
                return false
        }

        return verifyPassword(adminHash, pass)
}

func main() {
        var u, p string

        fmt.Print("Username: ")
        fmt.Scanln(&u)

        fmt.Print("Password: ")
        fmt.Scanln(&p)

        if login(u, p) {
                fmt.Println("Welcome Admin!")
        } else {
                fmt.Println("Access Denied")
        }
}
```